

StudyConnect End-to-End Test Plan

Oct 8, 2026 · @Omkar

This plan walks one new student from sign-up to profile, lab enrolment, Team Finder matching, team formation and real-time chat, then checks every other feature. Run it top to bottom in about 60 minutes and mark each case in the Result column.

Setup

Start from fresh demo data so the expected results below match exactly.

1. Stop the app and delete `src/StudyConnect.Web/App_Data` (or set `Database:ResetOnStartup` to `true` for one run).
2. Run **StudyConnect.Web** from Visual Studio 2022 (F5). The app opens at `http://localhost:5080` and seeds 30 demo students.
3. Use **two browsers**: a normal window (Browser A) and a private/incognito window (Browser B). Each window holds its own login, so you can watch real-time updates between two users.
4. Keep the browser console open (F12) and note any red errors in the defect log.

All demo passwords are `Demo@1234`.

Account	Role in the tests	Seeded facts
New user (you create it)	Main journey, sections 1–5	Created in TC-1.x, deleted in TC-13.x
alice@uts.edu.au	Second user for matching and chat	31927 Lab 02 (Mon 13:00–15:00, Mr. James Okafor) and DEMO-ML Lab A. Connected to Bob, Diana, Priya. Requests received from Charlie and Wendy, sent to Ethan. 2 unread messages. Invited to team LINQ Legends
bob@uts.edu.au	Lab 02 team mate	31927 Lab 02, Frontend, connected to Alice
george@uts.edu.au	Leader of LINQ Legends	31927 Lab 02, open team with 1 member
ethan@uts.edu.au	Different-lab student	31927 Lab 01, looking for a team

Account	Role in the tests	Seeded facts
noah@uts.edu.au	Leader of a full team	Null Pointers (Noah, Fatima, Isaac), Lab 01, full and closed
charlie@uts.edu.au	Password reset	31927 Lab 04

In 31927 Lab 02 the available students (not in a team, looking) are Bob, Diana, Priya, Wendy, Rahul and Zack; George already leads a team.

1. Account creation and login

Browser A. The new account used through sections 1–5 is **Sam Tester** / **sam.testers@uts.edu.au** / **Strong@123**.

ID	Steps	Expected result	Result
TC-1.1	Open <code>http://localhost:5080</code> while logged out	Redirected to <code>/account/login?ReturnUrl=%2F</code> ; "Welcome back" form with a demo-account hint	▼
TC-1.2	Click Create an account , then Create account with every field empty	Messages: "Please enter your name.", "Email is required.", "Password is required.", "Please confirm your password."	▼
TC-1.3	Type password <code>weak</code> and watch the rule list and strength meter	Rules tick live; meter reads Very weak; only "A lowercase letter" is ticked	▼
TC-1.4	Name <code>Sam Tester</code> , email <code>sam.testers@uts.edu.au</code> , contact <code>0400 000 000</code> , password <code>Strong@123</code> , confirm <code>Strong@124</code> , submit	"Passwords do not match." and no account is created	▼
TC-1.5	Email <code>not-an-email</code> , valid passwords, submit	"Invalid email format." (or the field validation message)	▼
TC-1.6	Email <code>alice@uts.edu.au</code> , valid passwords, submit	"Email already registered."	▼
TC-1.7	Correct the form (email <code>sam.testers@uts.edu.au</code> , both passwords <code>Strong@123</code>), submit	"Account created" then redirect to <code>/profile?welcome=true</code> showing "Welcome! Let's set up your profile"; the navbar shows Sam's avatar	▼

ID	Steps	Expected result	Result
TC-1.8	Log out (avatar menu), log in as <code>sam.testers@uts.edu.au</code> with <code>wrong@123</code>	"Invalid email or password."	▼
TC-1.9	Log in with <code>Strong@123</code> and tick Keep me signed in for 7 days	Dashboard loads with "Hi Sam" under a Good morning/afternoon/evening greeting; closing and reopening the browser keeps you logged in	▼
TC-1.10	While logged out, open <code>http://localhost:5080/team-finder</code> directly, then log in	After login you land on Team Finder (safe return URL); an external URL like <code>?ReturnUrl=https://evil.com</code> goes to the dashboard instead	▼

2. Profile setup

Browser A as Sam, on the welcome profile page from TC-1.7. Interests and free time feed the matching score, so fill them in carefully.

ID	Steps	Expected result	Result
TC-2.1	Click Save profile with no interests and no free time selected	"Please add at least one interest." / "Please select at least one time you're available."; nothing is saved	▼
TC-2.2	Under "Popular with other students" click + C# , + .NET , + Machine Learning	Each appears as a removable chip under Interests	▼
TC-2.3	Type <code>Robotics</code> in "Add your own..." and press Enter; then click × on Robotics	Chip is added (lower-cased/normalised), then removed	▼
TC-2.4	In "When are you free?" tap Monday Afternoon, Wednesday Morning, Friday Afternoon	Cells turn highlighted; tapping again toggles them off	▼
TC-2.5	Degree: start typing <code>Bachelor of I</code>	Auto-complete suggests "Bachelor of Information Technology"	▼
TC-2.6	Add a short bio and click Save profile	Toast "Profile saved ✓" and redirect to <code>/subjects?welcome=true</code> with the "Almost done!" banner	▼

ID	Steps	Expected result	Result
TC-2.7	Open Profile again from the avatar menu	Saved name, degree, bio, 3 interests and 3 free-time cells are shown; dashboard "Profile completeness" bar has risen	

3. Subject and lab enrolment

Browser A as Sam on **Subjects & labs**. Sam joins 31927 Lab 02 so he is matched with Alice's lab.

ID	Steps	Expected result	Result
TC-3.1	Look at the 31927 card	Code, name, "Teams of 2–3 students from the same lab" and a table of Lab 01–04 with day, time, room and tutor (Lab 02 = Mon 13:00–15:00, Mr. James Okafor)	▼
TC-3.2	Click + Enrol on 31927, leave the lab on "— Choose your lab —", click Save	Modal stays open with "Please choose your lab."	▼
TC-3.3	Paste a pitch longer than 300 characters and save	"Keep your pitch under 300 characters."	▼
TC-3.4	Lab 02, role Testing , confidence Intermediate , target High Distinction , switch "I'm looking for an assignment team" on, short pitch, Save	Toast "Enrolment saved"; card shows an Enrolled badge, tags for Lab 02 / Testing / Intermediate / High Distinction / "Looking for a team"; the Lab 02 row is highlighted	▼
TC-3.5	Click Edit preferences , change the role to Backend, save	Modal opens pre-filled with the saved values; tag updates to Backend	▼
TC-3.6	Enrol in DEMO-WEB (any lab), then click Unenrol on it and confirm	Confirm dialog "Unenrol from subject?"; toast "Unenrolled from DEMO-WEB"; card returns to + Enrol	▼
TC-3.7	On the 31927 card click 🔍 Team Finder	Opens <code>/team-finder?subject=<id></code> with 31927 already selected	▼

The rules that block unenrolling or changing labs while in a team are tested in section 5 (TC-5.10, TC-5.11).

4. Team Finder matching

Browser A as Sam on **Team Finder** with 31927 selected. Sam shares C#, .NET, Machine Learning, Monday afternoon and Wednesday morning with Alice Johnson, so Alice should rank at or near the top. Exact percentages depend on the profile you entered; check the order and the reasons, not fixed numbers.

ID	Steps	Expected result	Result
TC-4.1	Open Team Finder from the navbar	Subject picker set to 31927; header says matches come from your subject and your lab; "Your lab: Lab 02"; tabs Suggested teams / Students / Open teams	▼
TC-4.2	Open the Students tab	7 cards: Alice, Bob, Diana, Priya, Wendy, Rahul, Zack, sorted by score (highest first); George is hidden because he already leads a team	▼
TC-4.3	On Alice's card read the score ring and reasons, then expand Why this score?	High score with plain-language reasons (same lab, shared interests, overlapping free time); breakdown lists six criteria: same lab, shared interests, free-time overlap, complementary roles, goal alignment, skill balance	▼
TC-4.4	Drag the Min. match slider to 80%	Only students at or above 80% remain; the list updates instantly without a page reload; slide back to 0	▼
TC-4.5	Role filter Frontend	Only Bob and Wendy; set back to "Any role"	▼
TC-4.6	Type Blazor in Search	Students whose interests or pitch mention Blazor (e.g. Bob, Diana, Wendy)	▼
TC-4.7	Untick Only students without a team	George appears with an "In a team" badge	▼
TC-4.8	Untick Same lab only	Students from Lab 01/03/04 (e.g. Ethan) appear, each card showing their own lab; tick it again (the "Other lab" badge that replaces the invite button is checked in TC-5.9)	▼
TC-4.9	Open the Open teams tab	LINQ Legends (leader George, 1 member, Lab 02) with 🧑 Request to join ; Null Pointers (Lab 01, full) is not listed	▼

ID	Steps	Expected result	Result
TC-4.10	Open Suggested teams	Up to 3 complete teams of 3 (Sam + 2), each with a team score, average pair compatibility, roles covered and common free time; best first	▼

5. Teams

Browser A = Sam, Browser B = Alice (log in as `alice@uts.edu.au`). This is the core journey: matching turns into a real team with a live workspace.

ID	Steps	Expected result	Result
TC-5.1	Sam: on Suggested teams click Create team & invite on the suggestion that includes Alice	Toast "Team created"; redirect to <code>/teams/<id></code> ; Sam is listed as 🏰 Leader; both suggested students show as Pending invitations with a Cancel link	▼
TC-5.2	Alice (Browser B): watch the bell, then open My Teams	A live toast and a bell badge appear without refreshing; "✉️ Invitations for you" lists Sam's team and the older LINQ Legends invite	▼
TC-5.3	Alice: Accept Sam's invitation	Toast "Welcome to <team name>!"; Sam's workspace in Browser A adds Alice to Members instantly; LINQ Legends' invite to Alice is withdrawn (one team per subject)	▼
TC-5.4	Both: open the team workspace and check 📊 Team health	Average compatibility %, roles covered/missing, and a "When everyone is free" heat-map where Monday afternoon and Wednesday morning are darkest	▼
TC-5.5	Sam types <code>Hi team, lab 02 on Monday?</code> in 💬 Team chat and presses Enter; Alice replies	Each message appears in the other browser within about a second, with avatar and time	▼
TC-5.6	Sam: Cancel the remaining pending invitation	Toast "Invitation cancelled"; it disappears from Pending	▼

ID	Steps	Expected result	Result
TC-5.7	Sam:  Edit team → name A → Save; then name Lab 02 Legends , project idea, untick "Open to join requests", Save	First: "Team name must be between 3 and 60 characters."; second: toast "Team updated" and a "Closed" badge	▼
TC-5.8	Sam:  Export roster (CSV)	A CSV downloads with Name, Email, Contact, Role, Leader columns for Sam and Alice	▼
TC-5.9	Sam: Team Finder → Students, untick Same lab only	Lab 02 students show  Invite to team ; Ethan (Lab 01) shows an "Other lab" badge instead; tick it again	▼
TC-5.10	Sam: Subjects → 31927 → Unenrol	Error toast "Leave your team for this subject before unenrolling."	▼
TC-5.11	Sam: Subjects → 31927 → Edit preferences → change lab to Lab 03 → Save	"You are in a team for this subject. Leave the team before changing labs (teams must be from the same lab)."	▼
TC-5.12	Log in as ethan@uts.edu.au (Lab 01) in Browser B, Team Finder → Open teams	Sam's team is not listed (different lab, and closed); log back in as Alice	▼
TC-5.13	Join request: as Bob (Lab 02) open Team Finder → Open teams →  Request to join on LINQ Legends; then log in as George → My Teams → Approve	Bob sees "Request sent to LINQ Legends"; George sees "  Requests to join your team"; after Approve Bob is a member	▼
TC-5.14	Alice: team workspace → Leave team → confirm	Toast "You left the team"; Sam's Members list drops to 1 in real time; if the leader leaves, leadership passes to the longest-standing member	▼

After TC-5.14, have Sam invite Alice again from the workspace ( Invite a team mate) and accept it as Alice, so the team is intact for later sections.

6. Connections

Browser A = Sam, Browser B = Alice. Direct messages need an accepted connection, so this section sets up the chat in section 7.

ID	Steps	Expected result	Result
TC-6.1	Alice: open Connections	Tabs with counts: My connections 3 (Bob, Diana, Priya), Requests received 2 (Charlie, Wendy), Requests sent 1 (Ethan)	▼
TC-6.2	Sam: Team Finder → Students → Alice's card → + Connect	Toast "Connection request sent"; button changes to "⌚ Requested"	▼
TC-6.3	Alice: without refreshing, check the bell and Requests received	Bell badge increases and a toast appears; Requests received shows 3 including Sam Tester	▼
TC-6.4	Alice: ✓ Accept on Sam	Toast "You're now connected with Sam Tester"; Sam moves to My connections; Sam's card for Alice now shows 💬 Message	▼
TC-6.5	Alice: Decline Charlie; Requests sent → Cancel request on Ethan	Toasts "Request declined" and "Request cancelled"; both lists shrink	▼
TC-6.6	Alice: type Python in the search box on My connections	Only connections whose name or interests match remain	▼
TC-6.7	Alice: Remove Priya → confirm	Dialog "Remove connection?"; toast "Connection removed"; Priya leaves the list (her old conversation stays in Messages for TC-7.8)	▼

7. Real-time messaging

Browser A = Sam, Browser B = Alice, side by side. Every update in this section must appear without pressing F5.

ID	Steps	Expected result	Result
TC-7.1	Alice: open Messages	Conversations with Bob, Diana and Priya; Diana and Priya show unread counts; the navbar Messages pill shows 2	▼
TC-7.2	Alice: open Diana's conversation	Messages grouped under Today/Yesterday dividers; unread count clears and the navbar pill drops to 1	▼
TC-7.3	Sam: Connections → Alice →  Message	Opens <code>/messages/<Alice's id></code> with "No messages yet — say hello! 🙌"	▼
TC-7.4	Sam: click Send with an empty box	Nothing is sent (or "Message cannot be empty.")	▼
TC-7.5	Sam: type <code>Hey Alice, ready for Monday's lab?</code> and press Enter	Message appears for Sam with a single ✓ (Delivered); in Alice's window Sam's conversation jumps to the top with an unread count and the navbar pill increases, live	▼
TC-7.6	Alice: open Sam's conversation and reply	Sam's tick changes to ✓✓ ("Seen" on hover) live; Alice's reply appears in Sam's window within about a second	▼
TC-7.7	Alice: right-click Bob's conversation	Context menu with  Open chat, ✓ Mark as read and  Delete conversation; choose Delete conversation → confirm → toast "Conversation deleted"	▼
TC-7.8	Alice: open Priya's conversation (removed in TC-6.7) and send a message	Error "You must be connected to message this user."; nothing is stored	▼
TC-7.9	Alice: type <code>sam</code> in "Search conversations..."; then shrink the window below 700 px	List filters to Sam; on narrow screens the list and the chat show one at a time with a Back to conversations button	▼

8. Study sessions

Browser A = Sam, Browser B = Alice. Five upcoming sessions are seeded, including Alice's "AI Study Group" in 2 days.

ID	Steps	Expected result	Result
TC-8.1	Sam: open Sessions	Cards for AI Study Group, Blazor Components Workshop, EF Core & LINQ Revision, Data Science Discussion and NUnit Testing Clinic, each with time, duration, format, location and spots left	▼
TC-8.2	Sam: + New session , Starts = yesterday, other fields valid, Create session	"The session must start in the future."	▼
TC-8.3	Sam: title A , duration 5, capacity 1	"Title must be between 3 and 100 characters.", "Duration must be between 15 and 480 minutes.", "Capacity must be between 2 and 100."	▼
TC-8.4	Sam: title Lab 02 prep , topic Assignment 2 planning , subject 31927, tomorrow 10:00, 60 min, Online, a meeting link, capacity 5, Create session	Toast "Session created"; the card appears for Sam and, live, in Alice's Sessions page	▼
TC-8.5	Alice: Join session on "Lab 02 prep"	Toast that she joined; button becomes "✓ Going"; spots left drops to 3; Sam gets a bell notification that Alice joined	▼
TC-8.6	Alice: Leave the session; Sam: try to leave his own session	Alice: "You left the session". Sam: only Delete is offered (a creator cannot leave their own session)	▼
TC-8.7	Sam: filters — subject 31927, search LINQ , "Only sessions I'm attending", "Show past sessions"	List narrows as each filter is applied, no page reload	▼
TC-8.8	Sam: Delete "Lab 02 prep" → confirm	Dialog "Delete session?"; toast "Session deleted"; card disappears in both browsers	▼

9. Notifications

The bell collects everything the earlier sections triggered: connection requests and acceptances, team invitations, join requests, team updates and session joins.

ID	Steps	Expected result	Result
TC-9.1	Sam: click the  bell	Dropdown lists the notifications from sections 5–8, newest first, with relative times ("5 minutes ago"); unread ones are highlighted; badge shows the unread count (9+ above nine)	▼
TC-9.2	Sam: click a notification (e.g. Alice joined your session)	The menu closes and you are taken to the related page (here, Sessions)	▼
TC-9.3	Sam: Mark all as read	Badge disappears, highlighting is removed and the "Mark all as read" link hides	▼
TC-9.4	While Sam is on the Dashboard, Alice triggers an event for him (e.g. joins a session Sam created)	A toast slides in on Sam's screen and the badge increments without a refresh	▼

10. Insights and machine learning

Browser A as Sam on **Insights**. This screen carries the machine-learning bonus, so show it in the demo.

ID	Steps	Expected result	Result
TC-10.1	Open Insights with "All subjects"	Stat tiles: Students (31 with Sam), Looking for a team, Teams formed, Upcoming sessions, Online right now	▼
TC-10.2	Hover and click bars in each chart	Four bar charts: Most popular interests, Students still needing a team per lab (labels include the subject code, e.g. "31927 Lab 02"), Preferred team roles, Free time slots by day; clicking a bar shows its detail line with a dismiss link	▼

ID	Steps	Expected result	Result
TC-10.3	Change Subject to 31927	Lab chart switches to "Lab 01 (Mon)" style labels; Lab 02's bar is lower than before section 5 because Sam and Alice are now in a team	▼
TC-10.4	Scroll to 🧠 Study communities (K-Means clustering)	Several community cards, each named by its top interests (e.g. "Cloud Computing · Cyber Security · Python", "SQL · C# · .NET") with member names; Sam is listed in the community whose label best matches his interests	▼
TC-10.5	Expand How does this work?	Explains TF-IDF feature extraction, K-Means with k-means++ and $k \approx \sqrt{n/2}$, and centroid labelling	▼

11. Password reset and security

Use a fresh private window (Browser C) for the reset; Charlie's account is used so Sam and Alice stay untouched.

ID	Steps	Expected result	Result
TC-11.1	Login page → Forgot password? → email <code>charlie@uts.edu.au</code> → Send reset link	"Check your email" page; in Development mode an Open reset link button is shown and an HTML e-mail is written to <code>App_Data/outbox</code>	▼
TC-11.2	Forgot password with <code>nobody@uts.edu.au</code>	"If an account exists for that e-mail, a reset link has been sent." with no reset button (the app does not reveal which e-mails exist)	▼
TC-11.3	Click Open reset link , enter <code>NewPass@2026</code> and <code>NewPass@2027</code>	"Passwords do not match."	▼
TC-11.4	Enter <code>NewPass@2026</code> twice, Reset password	"Password reset successful" with Go to login ; logging in as Charlie with <code>NewPass@2026</code> works and <code>Demo@1234</code> no longer does	▼

ID	Steps	Expected result	Result
TC-11.5	Open the same reset link again	"Invalid or expired reset link. Please request a new one." (tokens are single-use and expire after 30 minutes)	▼
TC-11.6	Open <code>/account/reset-password</code> with no token	"Invalid link — This password reset link is missing its token or has been mangled."	▼
TC-11.7	Sam: Profile → Security tab → wrong current password → new <code>Strong@456</code> twice → Update password ; then the correct current password	First: "Your current password is incorrect."; second: toast "Password updated"	▼
TC-11.8	Log out, then press the browser Back button and open <code>/messages</code>	Redirected to login; no private page is shown from cache	▼

Optional checks for the report: open `App_Data/studyconnect.db` in DB Browser for SQLite and confirm the Users table stores PBKDF2 hashes, never plain passwords, and the PasswordResetTokenHash column holds only a SHA-256 hash.

12. Theme, responsive layout and error handling

ID	Steps	Expected result	Result
TC-12.1	Click the  theme button in the navbar, then reload the page	Whole app switches to dark mode; the choice survives a reload (saved in the browser)	▼
TC-12.2	Open DevTools device mode at 375 px wide (iPhone) and visit Dashboard, Team Finder, Team workspace, Messages	Navbar collapses to a menu button; cards stack in one column; no horizontal scrolling	▼
TC-12.3	Open <code>/teams/99999</code>	"Team not found — This team may have been deleted." with a link back	▼
TC-12.4	As Ethan, open the URL of Sam's team workspace	"Only team members can see the team workspace."	▼

ID	Steps	Expected result	Result
TC-12.5	Open <code>/does-not-exist</code>	Logged in: "Page not found" inside the app layout with Back to dashboard. Logged out: redirect to the login page	▼
TC-12.6	Stop the app in Visual Studio while a page is open, then restart it	Blazor shows its reconnect banner, then reconnects (or offers Reload) once the app is back	▼

13. Logout and account deletion

Run this last: it removes Sam and everything he created. Keep Alice logged in to Browser B.

ID	Steps	Expected result	Result
TC-13.1	Sam: avatar menu → Log out	Login page with "You have been logged out. See you soon!"; Back button does not reopen private pages	▼
TC-13.2	Log in as Sam again (<code>Strong@456</code> after TC-11.7) → avatar menu → My profile → Delete account tab → Delete my account	Dialog "Delete your account?" warning "This cannot be undone."	▼
TC-13.3	Click Yes, delete everything	Redirect to <code>/account/login?deleted=true</code> with "Your account and data have been deleted."	▼
TC-13.4	Alice (Browser B): reload My Teams, Connections, Messages and Team Finder	Sam is gone everywhere; Alice is now 🏰 Leader of the team (leadership passes to the longest-standing member); Sam's conversation is removed	▼
TC-13.5	Try to log in as <code>sam.test@uts.edu.au</code>	"Invalid email or password."; signing up again with the same e-mail is allowed	▼

14. Automated NUnit tests

The 64 NUnit test cases back up the manual run: they check the same rules without a browser.

1. In Visual Studio open **Test → Test Explorer** and click **Run All Tests** (or run `dotnet test` in the solution folder).
2. Expect 64 passed, 0 failed. Integration tests use an in-memory SQLite database, so they never touch `App_Data`.
3. Take a screenshot of Test Explorer for the report.

Fixture	Cases	What it proves
ExtensionMethodTests	14	Tag normalising, title case, initials, relative time, display names
PasswordPolicyTests	9	Each password rule and the strength score
MachineLearningTests	7	TF-IDF weights, cosine similarity, K-Means and k-means++ group similar students
CommonTypeTests	6	<code>Result<T></code> , combinations iterator, weekly availability enumerator and indexer
CompatibilityEngineTests	6	Same-lab, interests, free time, roles, goals and skill criteria and their weighting
AuthServiceTests	5	Register → login, duplicate e-mail / weak password / mismatch, wrong password, full reset flow, unknown e-mail reveals nothing
PasswordHasherTests	4	PBKDF2 hashes are salted and verify correctly
TeamBuilderTests	4	Suggested teams have the right size, are best-first and cover roles
ConnectionAndMessageServiceTests	4	Messaging needs an accepted connection; unread counts; a request sent back accepts the existing one; real-time subscribers receive new messages
TeamServiceTests	4	Same-lab and team-size rules, one team per subject, leader hand-over, Team Finder only suggests students from my lab

Fixture	Cases	What it proves
DatabaseSeederTests	1	JSON seed files load from disk into the database

ID	Steps	Expected result	Result
TC-14.1	Run All Tests in Test Explorer	64 passed, 0 failed, 0 skipped	▼

Sign-off

Item	Value
Tester	
Date run	
Build / commit	
Browsers	
Cases passed / total	/ 101

Defect log — one row per failed case; attach a screenshot and the browser console output.

Case ID	What happened	Severity	Screenshot	Fixed in

All 101 cases passing (100 manual plus TC-14.1, which runs the 64 automated tests) is the bar for the demo build.